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Homework 12

1.

We are given the stochastic differential equation for the Ornstein-Uhlenbrek process:

$$dX(t) = \theta(\mu - X(t))dt + \sigma dW(t)$$

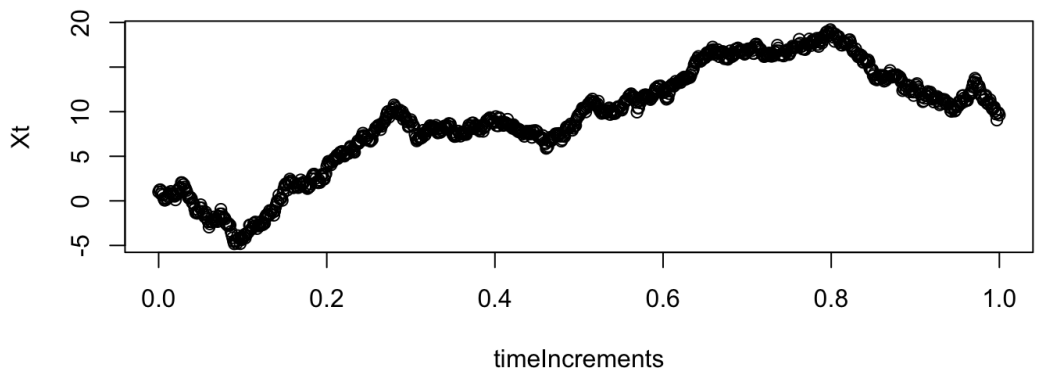
with the analytical solution as

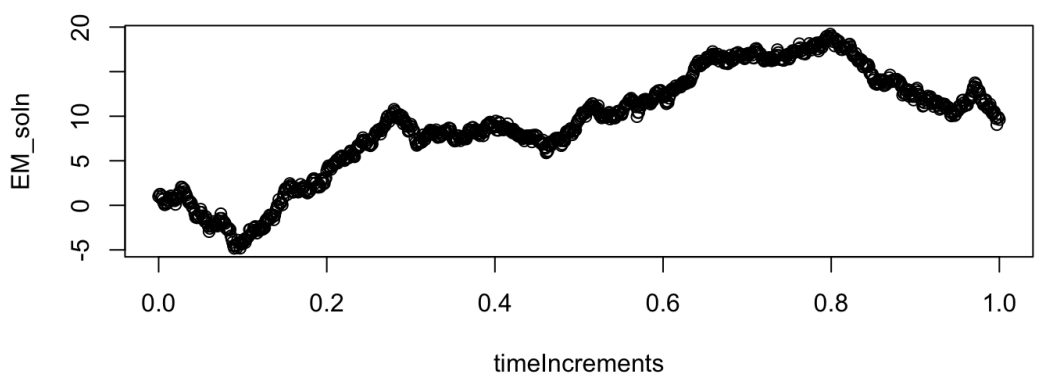
$$X(t) = X(0)e^{-\theta t} + \mu(1 - e^{-\theta t}) + \int_0^t \sigma e^{\theta(s-t)} dW(s)$$

with convenient parameters as $\theta = 1, \mu = 20, \sigma = 10$. We aim to compare this with the solution found using the Euler-Maruyama approximation. Let's simulate and compare:

```
> T = 1
> N = 1000
> theta = 1
> mu = 20
> sigma = 10
>
> ornstein_uhlenbrek = function(T, N, theta, mu, sigma) {
+
+   # loop over time steps
+   dT = T / N
+   timeIncrements = seq(0, T, dT)
+   W_vec = rep(0, N+1)
+   Xt = rep(0, N+1)
+   X0 = 1
+   Xt[1] = X0
+   EM_soln = rep(0, N+1)
+   EM_soln[1] = X0
+   stochastic_integral = 0
+
+   for (j in 1:N) {
```

```
+     dW = rnorm(1, mean = 0, sd = sqrt(dT))
+     W_vec[j+1] = W_vec[j] + dW
+
+     # EM approximation for OU process
+     EM = EM_soln[j] + theta * (mu - EM_soln[j]) * dT + sigma * dW
+     EM_soln[j+1] = EM
+
+     # Calculate exact solution;
+     # Taking the full stochastic integral
+     s = timeIncrements[j]
+     t = timeIncrements[j+1]
+     stochastic_integral = stochastic_integral + sigma * exp(theta * s)
* dW
+
+     constant_part = X0 * exp(-theta * t) + mu * (1 - exp(-theta * t))
+
+     Xt[j+1] = constant_part + stochastic_integral * exp(-theta * t)
+   }
+
+   # Plot results
+   par(mfrow = c(2,1));
+   plot(timeIncrements, Xt);
+   plot(timeIncrements, EM_soln);
+ }
>
> ornstein_uhlenbrek(T, N, theta, mu, sigma)
```





We see that the comparison shows that the Euler-Maruyama approximation works very well here and basically exactly matches the actual exact solution.

2.

We aim to illustrate through R the difference between the Ito ingeral and the Stratonovich integral; i.e. find some value of $h(t)$ and simulated process $W(t)$ so that calculating the integral $\int_0^T h(t)dW(t)$ is considerably different for the two calculations.

As explained in the module, the core difference between the Ito and Stratonovich is that we evaluate W at the left endpoint compared to the middle point of each interval when integrating. We can use the same example provided, i.e. setting $h(t) = W(t)$.

We know that the resulting integrals are $\frac{1}{2}W(T)^2 - \frac{1}{2}T$ and $\frac{1}{2}W(T)^2$ respectively, so theoretically we expect a difference of $\frac{1}{2}T$ between the values, which is quite large as a term. Performing the simulation:

```
> T_values = c(0.5, 1, 5)
> N = 1000
> nmbSimulations = 1000
>
> ito_strat_comparison = function(T, N, nmbSimulations) {
+
+   ito_integrals = rep(0, nmbSimulations)
+   strat_integrals = rep(0, nmbSimulations)
+   W_final_squared = rep(0, nmbSimulations)
+
+   for (iSim in 1:nmbSimulations) {
+     dT = T / N
+     timeIncrements = seq(0, T, dT)
+
+     # Initialize Brownian motion
+     W_values = rep(0, N+1)
+     W_values[1] = 0
+
+     # Generate Brownian motion path
+     for (j in 1:N) {
+       dW = rnorm(1, 0, sqrt(dT))
+       W_values[j+1] = W_values[j] + dW
+     }
+
+     ito_integral = 0
+     strat_integral = 0
+     for (j in 1:N) {
```

```
+       # Ito integral (just left endpoint)
+       dW = W_values[j+1] - W_values[j]
+       ito_integral = ito_integral + W_values[j] * dW
+
+       # Stratonovich integral (calculate midpoint with delta Z adjustment)
+       delta_z = rnorm(1, mean = 0, sd = sqrt(dT / 4))
+       W_midpoint = 0.5 * (W_values[j] + W_values[j+1]) + delta_z
+       strat_integral = strat_integral + W_midpoint * dW
+     }
+
+     ito_integrals[iSim] = ito_integral
+     strat_integrals[iSim] = strat_integral
+     W_final_squared[iSim] = W_values[N+1]^2
+   }
+
+   # Compute averages to compare values
+   ito_int_avg = mean(ito_integrals)
+   strat_int_avg = mean(strat_integrals)
+   avg_W_squared = mean(W_final_squared)
+
+   # Calculate theoretical values to compare
+   ito_int_theoretical = 0.5 * avg_W_squared - 0.5 * T
+   strat_int_theoretical = 0.5 * avg_W_squared
+
+   sprintf("Results for T = %f, N = %d...", T, N)
+
+   print("Ito Integral Results")
+   print(paste("Simulated average:", ito_int_avg))
+   print(paste("Theoretical value:", ito_int_theoretical))
+   print(paste("Simulated - theoretical difference:", ito_int_avg - ito_int
```

```
+
+   print("Ito Integral Results")
+   print(paste("Simulated average:", strat_int_avg))
+   print(paste("Theoretical value:", strat_int_theoretical))
+   print(paste("Simulated - theoretical difference:", strat_int_avg -
strat_int_theoretical))
+
+   print("Ito-Stratanovich Difference Results")
+   print(paste("Simulated difference:", ito_int_avg - strat_int_avg))
+   print(paste("Theoretical difference:", ito_int_theoretical - strat_int_t
+ }
>
> for (T in T_values) {
+   ito_strat_comparison(T, N, nmbSimulations)
+ }
[1] "Ito Integral Results"
[1] "Simulated average: -0.00042617134466108"
[1] "Theoretical value: -0.00156682061385677"
[1] "Simulated - theoretical difference: 0.00114064926919569"
[1] "Ito Integral Results"
[1] "Simulated average: 0.248481126129809"
[1] "Theoretical value: 0.248433179386143"
[1] "Simulated - theoretical difference: 4.79467436660941e-05"
[1] "Ito-Stratanovich Difference Results"
[1] "Simulated difference: -0.24890729747447"
[1] "Theoretical difference: -0.25"
[1] "Ito Integral Results"
[1] "Simulated average: 0.00570368151589045"
[1] "Theoretical value: 0.00634295358654036"
[1] "Simulated - theoretical difference: -0.000639272070649901"
```

```
[1] "Ito Integral Results"
[1] "Simulated average: 0.506643545799045"
[1] "Theoretical value: 0.50634295358654"
[1] "Simulated - theoretical difference: 0.000300592212504669"
[1] "Ito-Stratanovich Difference Results"
[1] "Simulated difference: -0.500939864283155"
[1] "Theoretical difference: -0.5"
[1] "Ito Integral Results"
[1] "Simulated average: 0.146082889476072"
[1] "Theoretical value: 0.14837836668234"
[1] "Simulated - theoretical difference: -0.00229547720626805"
[1] "Ito Integral Results"
[1] "Simulated average: 2.64776303053536"
[1] "Theoretical value: 2.64837836668234"
[1] "Simulated - theoretical difference: -0.000615336146977175"
[1] "Ito-Stratanovich Difference Results"
[1] "Simulated difference: -2.50168014105929"
[1] "Theoretical difference: -2.5"
```

We see that th average simulated difference is indeed considerably different, and matches consistently with the theoretical difference as a constant of the T value.

3.

The approximation given in (5.4) of the paper is

$$\log e_{\Delta t}^{\text{strong}} \approx \log C + \frac{1}{2} \log \Delta t$$

so we try to illustrate this by plotting the resulting graph of average error vs. Δt on a log scale, where we should see an approximately straight line with a slope of $\frac{1}{2}$. We modify the function slightly to return $averageFinalDifference$, and then run accordingly and graph the results:

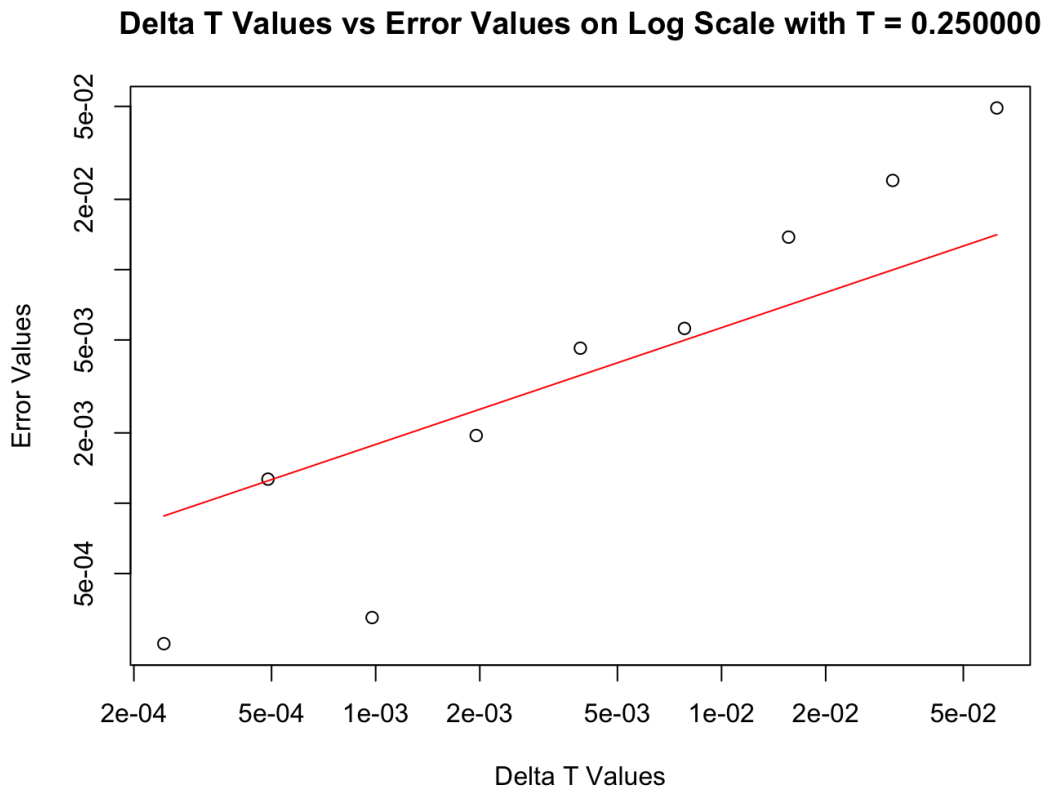
```
> T_values = c(0.25, 0.5, 1, 2, 5, 10)
```

```
> N_values = 2^(2:10)
> # Same as paper
> lambda = 2
> mu = 1
> nmbSimulations = 1000
>
> for (T in T_values) {
+   deltat_values = as.numeric(list())
+   error_values = as.numeric(list())
+
+   for (N in N_values) {
+     sprintf("running for T = %f, N = %d...", T, N)
+
+     deltat_values = append(deltat_values, T / N)
+     error_values = append(error_values, abs(blackScholesConvergence(T,
N, lambda, mu, nmbSimulations)))
+   }
+
+   plot(
+     deltat_values,
+     error_values,
+     log = "xy",
+     xlab = "Delta T Values",
+     ylab = "Error Values",
+     main = sprintf("Delta T Values vs Error Values on Log Scale with
T = %f", T)
+   )
+
+   # Reference line with exact slope as 1/2 fitted to similar offset
+   ref_C = exp(mean(log(error_values)) - 0.5 * mean(log(deltat_values)))
```

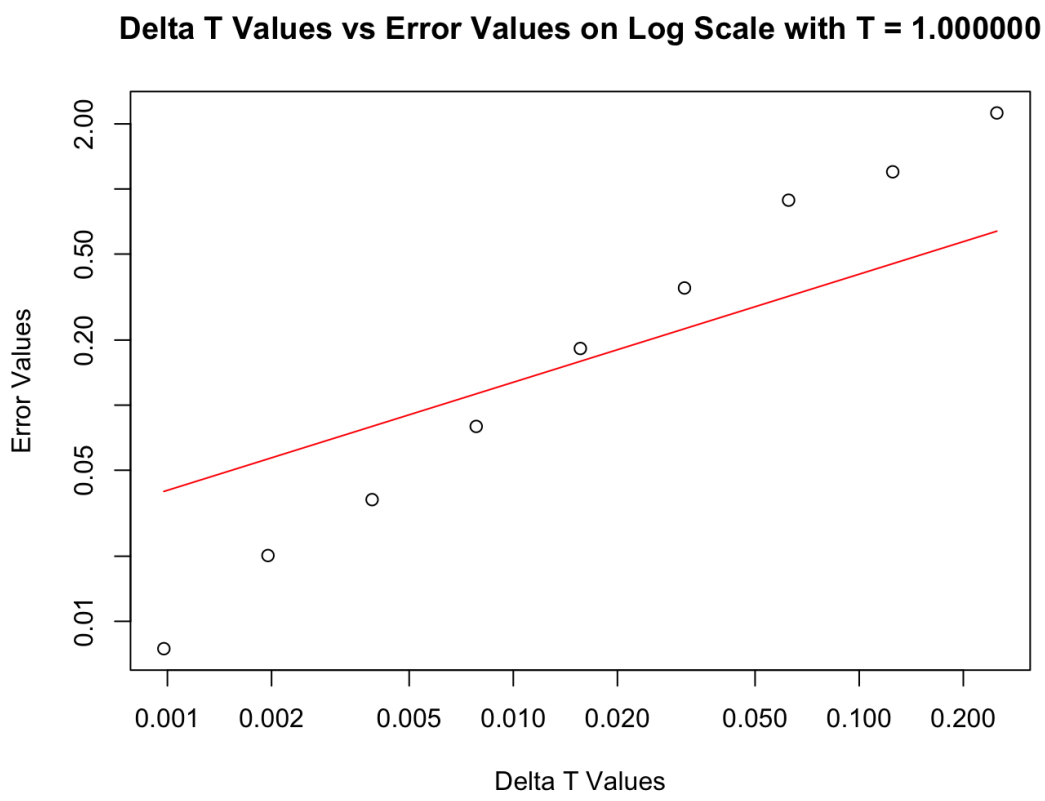
```
+   y = ref_C * sqrt(deltat_values)
+   lines(
+     deltat_values,
+     y,
+     type="l",
+     col = "red",
+   )
+ }
[1] "average deviation =  0.0492247166559775 , and the delta t =  0.0625"
[1] "average deviation =  0.024091028137924 , and the delta t =  0.03125"
[1] "average deviation =  0.0137789607057976 , and the delta t =  0.015625"
[1] "average deviation =  0.00560154546568383 , and the delta t =  0.0078125
[1] "average deviation =  0.00461061928605198 , and the delta t =  0.0039062
[1] "average deviation =  0.00194927934579275 , and the delta t =  0.0019531
[1] "average deviation =  0.000323983109936274 , and the delta t =  0.000976
[1] "average deviation =  0.00126783021087313 , and the delta t =  0.0004882
[1] "average deviation =  0.000250399393489314 , and the delta t =  0.00024
[1] "average deviation =  0.31711713984836 , and the delta t =  0.125"
[1] "average deviation =  0.132171536010918 , and the delta t =  0.0625"
[1] "average deviation =  0.0743188255687851 , and the delta t =  0.03125"
[1] "average deviation =  0.051269002108907 , and the delta t =  0.015625"
[1] "average deviation =  0.0197699304179946 , and the delta t =  0.0078125"
[1] "average deviation =  0.0150187575230333 , and the delta t =  0.00390625
[1] "average deviation =  0.00449707590957639 , and the delta t =  0.0019531
[1] "average deviation =  0.00146653530308295 , and the delta t =  0.000976
[1] "average deviation =  0.000595854856125777 , and the delta t =  0.00048
[1] "average deviation =  2.2471645677081 , and the delta t =  0.25"
[1] "average deviation =  1.20059710406355 , and the delta t =  0.125"
[1] "average deviation =  0.887552190793862 , and the delta t =  0.0625"
[1] "average deviation =  0.348304934738151 , and the delta t =  0.03125"
```

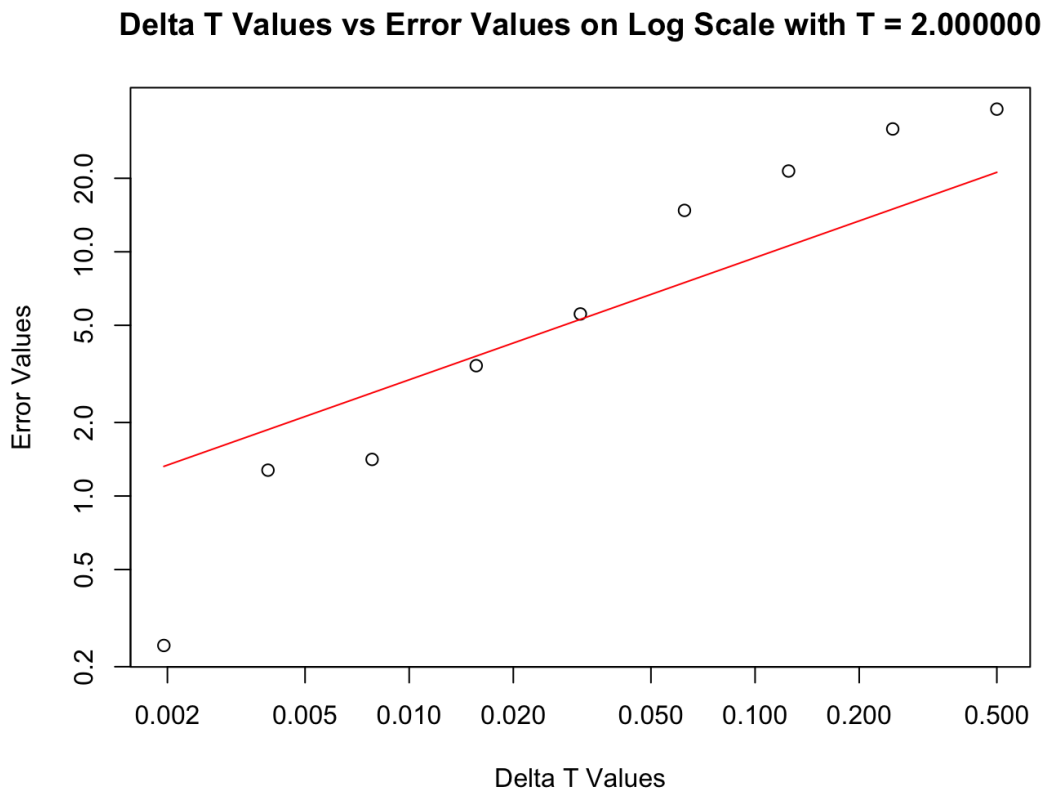
```
[1] "average deviation =  0.182839774132024 , and the delta t =  0.015625"
[1] "average deviation =  0.0797147803555463 , and the delta t =  0.0078125"
[1] "average deviation =  0.0365496282298914 , and the delta t =  0.00390625
[1] "average deviation =  0.0201513669795156 , and the delta t =  0.00195312
[1] "average deviation =  0.00747017151247117 , and the delta t =  0.0009765
[1] "average deviation =  38.3860350148055 , and the delta t =  0.5"
[1] "average deviation =  31.8416907859503 , and the delta t =  0.25"
[1] "average deviation =  21.4146284433919 , and the delta t =  0.125"
[1] "average deviation =  14.7718342858946 , and the delta t =  0.0625"
[1] "average deviation =  5.56539348123626 , and the delta t =  0.03125"
[1] "average deviation =  3.41716607124377 , and the delta t =  0.015625"
[1] "average deviation =  1.41197509467434 , and the delta t =  0.0078125"
[1] "average deviation =  1.2757240639073 , and the delta t =  0.00390625"
[1] "average deviation =  0.24428420763604 , and the delta t =  0.001953125"
[1] "average deviation =  15597.69661595 , and the delta t =  1.25"
[1] "average deviation =  15255.3191487282 , and the delta t =  0.625"
[1] "average deviation =  20462.1594433599 , and the delta t =  0.3125"
[1] "average deviation =  20943.8653605296 , and the delta t =  0.15625"
[1] "average deviation =  9256.47853698749 , and the delta t =  0.078125"
[1] "average deviation =  5654.01247705172 , and the delta t =  0.0390625"
[1] "average deviation =  2366.18215557262 , and the delta t =  0.01953125"
[1] "average deviation =  2552.93685449814 , and the delta t =  0.009765625"
[1] "average deviation =  788.906730669276 , and the delta t =  0.0048828125
[1] "average deviation =  287769024.071099 , and the delta t =  2.5"
[1] "average deviation =  513211342.45659 , and the delta t =  1.25"
[1] "average deviation =  362676431.210016 , and the delta t =  0.625"
[1] "average deviation =  364272514.331549 , and the delta t =  0.3125"
[1] "average deviation =  265851877.484744 , and the delta t =  0.15625"
[1] "average deviation =  229631572.665997 , and the delta t =  0.078125"
[1] "average deviation =  161346784.854953 , and the delta t =  0.0390625"
```

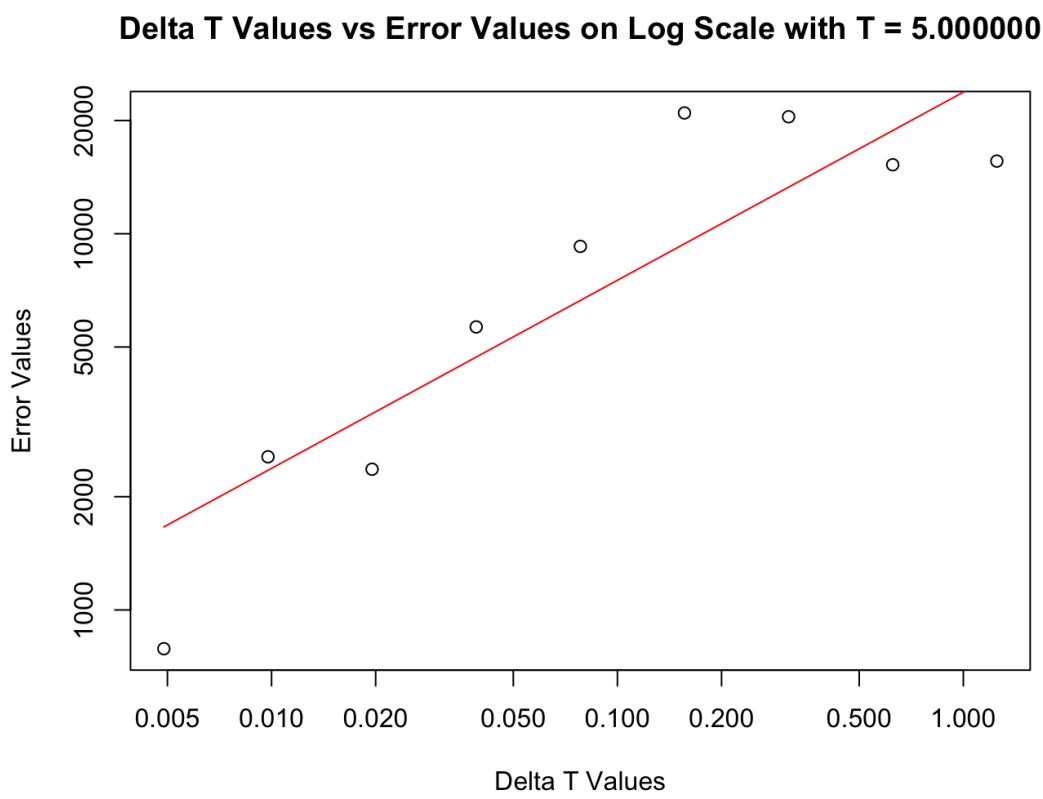
```
[1] "average deviation =  254194191.496466 , and the delta t =  0.01953125"
[1] "average deviation =  36543557.0686745 , and the delta t =  0.009765625"
```

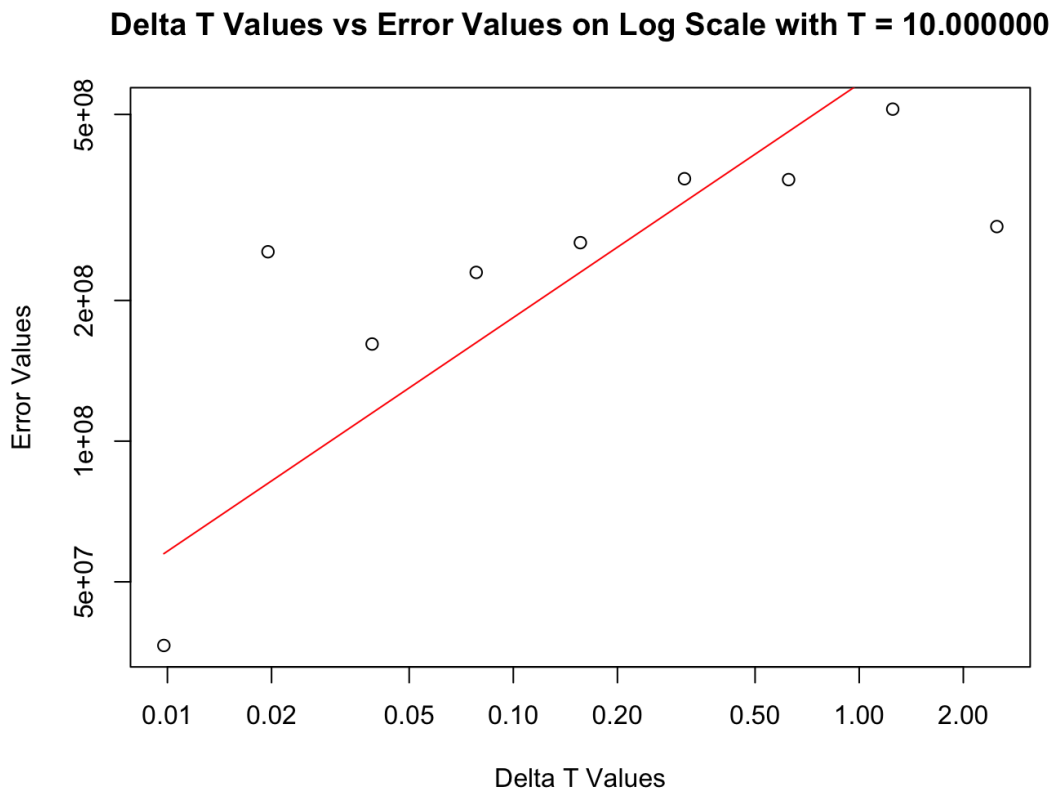


Delta T Values vs Error Values on Log Scale with T = 0.500000
Error Values
5e-04
2e-03
1e-02
5e-02
2e-01
5e-04
1e-03
2e-03
5e-03
1e-02
2e-02
5e-02
1e-01
Delta T Values


Delta T Values vs Error Values on Log Scale with T = 1.000000
Error Values
0.01
0.05
0.20
0.50
2.00
0.001
0.002
0.005
0.010
0.020
0.050
0.100
0.200
Delta T Values


Delta T Values vs Error Values on Log Scale with T = 2.000000
Error Values
0.2
0.5
1.0
2.0
5.0
10.0
20.0
0.002
0.005
0.010
0.020
0.050
0.100
0.200
0.500
Delta T Values


Delta T Values vs Error Values on Log Scale with T = 5.000000
Error Values
1000
2000
5000
10000
20000
0.005
0.010
0.020
0.050
0.100
0.200
0.500
1.000
Delta T Values



We plotted the adjusted lines with slope of $\frac{1}{2}$ and we do see that generally, the results do follow the approximation given in the paper. At $T = 10$, we begin to falter a bit more, but it is pretty close for all the other values for T.